

Christina Giannoula

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University of Toronto
40 St. George Street, M5S2E4
Toronto, Canada

EDUCATION

National Technical University of Athens (NTUA) **PhD** in Electrical and Computer Engineering
Advisors: Prof. [Georgios Goumas](#) (primary), Prof. [Nectarios Koziris](#), Prof. [Onur Mutlu](#) Nov 2016–Oct 2022
PhD Thesis: *Accelerating Irregular Applications via Efficient Synchronization and Data Access Techniques*
The PhD thesis is publicly available at <https://arxiv.org/abs/2211.05908>.
The PhD thesis has received the [Iakovos Giurunlian award for the best doctoral thesis with industrial interest](#).
The announcement is available [here](#).

National Technical University of Athens (NTUA) **Diploma** in Electrical and Computer Engineering
5-year Degree, Masters equivalent, GPA: 9.30/10.0 (top 2%) Nov 2011–Oct 2016
Advisor: [Prof. Georgios Goumas](#)
Diploma Thesis: *Parallelization Techniques for Concurrent Data Structures and Graph Algorithms* ([Grade 10/10](#))

PROFESSIONAL EXPERIENCE

University of Toronto (UofT) Toronto, Canada
[Postdoctoral Researcher](#) Jan 2023–Present

Advisor: [Prof. Gennady Pekhimenko](#)
Position Description: Postdoctoral researcher working and collaborating with [Prof. Gennady Pekhimenko](#), [Prof. Andreas Moshovos](#), [Prof. Nandita Vijaykumar](#) and their research groups.

CentML, [centml.ai](#) Toronto, Canada
[Postdoctoral Researcher](#) Feb 2024–Jun 2025

Position Description: Postdoctoral researcher working with research engineers in the field of systems for machine learning.

ETH Zürich Remote
[Senior Researcher](#) Jan 2020–Present

Position Description: Research collaborator with the SAFARI research group led by [Prof. Onur Mutlu](#).

EuroCC ◦ EU-funded project Athens, Greece
[Research Assistant](#) Apr 2022–Aug 2022

Position Description: Coordinate activities and perform training sessions in HPC-related fields and technologies (www.eurocc-access.eu/).

Cybele ◦ EU-funded project Athens, Greece
[Software Engineer](#) May 2020–Apr 2022

Project Description: Fostering precision agriculture and livestock farming through secure access to large-scale HPC-enabled virtual industrial experimentation environment empowering scalable big data analytics (www.cybele-project.eu).

ETH Zürich Zürich, Switzerland
[PhD Researcher](#), Advisor: [Prof. Onur Mutlu](#) Jan 2019–Dec 2019

Position Description: Visiting PhD researcher at the Systems Group (SAFARI research group) in the Computer Science Department.

Bonseyes ◦ **EU-funded project**
Software Engineer

Athens, Greece
Jun 2017–Jan 2019

Project Description: Transforming AI development from a cloud centric model, dominated by large internet companies, to an edge device centric model through a marketplace and an open AI platform (www.bonseyes.eu).

AWARDS AND GRANTS

EECS Rising Star October 2024
Awarded as Rising Star in Electrical Engineering and Computer Science (EECS) for the year of 2024 (70 selected from 370 applicants, **18.9%** selection rate).

Machine Learning and Systems (MLSys) Rising Star July 2024
Awarded as MLCommons Rising Star in Machine Learning and Systems for the year of 2024 (41 selected from 170 applicants, **24.1%** selection rate).

PostDoc Research Grant August 2023
Postdoctoral Research Grant for the years of 2022-2023, 2023-2024 and 2024-2025 from the Vector Institute for Artificial Intelligence (<https://vectorinstitute.ai/>), **CAD15000**.

Doctoral Thesis Award August 2023
Iakovos Giurunlian Award for the Best PhD/Doctoral Thesis with a subject of Industrial Interest in the year of 2022 at NTUA, **€1750**.

PhD Fellowship Sep 2021–Oct 2022
Funded Research Fellowship from the Foundation for Education and European Culture (IPEP), **€4000**.

PhD Fellowship Aug 2017–Apr 2020
Funded Research Fellowship from the General Secretariat for Research and Technology (GSRT) and the Hellenic Foundation for Research and Innovation (HFRI), **€27900**.

Distinction for Academic Excellence 2014–2015
“Paris Kanellakis” Award at ECE NTUA for the Highest GPA over all Courses in the Field of Information Technology (**GPA 9.67/10**), **€2400**.

Other Research Awards/Grants

- Greek Women in STEM Award in Technology Field, September 2024, **€200**.
- Thomaidion Award at NTUA for the Research Paper “DaeMon: Architectural Support for Efficient Data Movement in Fully Disaggregated Systems” published at SIGMETRICS 2023, August 2024, **€270**.
- Thomaidion Award at NTUA for the Research Paper “High-Performance and Balanced Parallel Graph Coloring on Multicore Platforms” published at Journal of Supercomputing 2022, July 2023, **€360**.
- Thomaidion Award at NTUA for the Research Paper “SynCron: Efficient Synchronization Support for Near-Data-Processing Architectures” published at HPCA 2021, November 2022, **€480**.
- Student Travel Grant Award to Attend the ACM SIGMETRICS Conference 2022 in Mumbai, June 2022, **\$900**.
- Thomaidion Award at NTUA for the Research Paper “NATSA: A Near-Data Processing Accelerator for Time Series Analysis” published at ICCD 2020, May 2022, **€360**.
- HiPEAC Paper Award for the Research Paper “SynCron: Efficient Synchronization Support for Near-Data-Processing Architectures” published at HPCA 2021, March 2021.
- Thomaidion Award at NTUA for the Research Paper “Combining HTM with RCU to Speed Up Graph Coloring on Multicore Platforms” published at ISC 2018, March 2021, **€320**.

- Student Travel Grant Award to Attend the ACM PACT Conference 2018 in Limassol, November 2018, \$500.
- 2nd Place Winner at PACT ACM Student Research Competition (SRC), November 2018.
- Student Travel Grant Award to Attend the HiPEAC ACACES Summer School 2018 in Fiuggi, July 2018 €1000.
- Student Travel Grant Award to Attend the IEEE HPCA Conference 2018 in Vienna, February 2018, \$800.
- Student Travel Grant Award to Attend the HiPEAC ACACES Summer School 2017 in Fiuggi, July 2017 €1000.
- “Antonios Sarafis” Award at ECE NTUA for the Highest GPA of the Academic Year (GPA 9.67/10), 2014-2015, €420.
- “Paris Kanellakis” Award at ECE NTUA for the Highest GPA over all Courses in the Field of Information Technology (GPA 8.79/10), 2013-2014, €580.

PUBLICATIONS

24. Konstantinos-Nikolaos Papadopoulos, **Christina Giannoula**, Nikolaos-Charalampos Papadopoulos, Nektarios Koziris, José M.G. Merayo, Dionisios N. Pnevmatikatos
Evaluating Microarchitectural Hardware Fault Detection for Application-Specific Requirements
Proceedings of the International Conference on Embedded Computer Systems: Architectures, Modeling and Simulation (**SAMOS**), 2025.
23. **Christina Giannoula**, Peiming Yang, Ivan Fernandez Vega, Jiacheng Yang, Sankeerth Durvasula, Yu Xin Li, Mohammad Sadrosadati, Juan Gomez Luna, Onur Mutlu, Gennady Pekhimenko
PyGim: An Efficient Graph Neural Network Library for Real Processing-In-Memory Architectures
Proceedings of the ACM International Conference on Measurement and Modeling of Computer Systems (**SIGMETRICS & POMACS**), 2025.
22. Qidong Su, Wei Zhao, Xin Li, Muralidhar Andoorveedu, Chenhao Jiang, Zhanda Zhu, Kevin Song, **Christina Giannoula**, Gennady Pekhimenko
Seesaw: High-throughput LLM Inference via Model Re-Sharding
Proceedings of the 8th Annual Conference on Machine Learning and Systems (**MLSys**), 2025.
Officially artifact evaluated as available and functional.
Outstanding Paper Honorable Mention at MLSys 2025.
21. Pawan Kumar Sanjaya, **Christina Giannoula**, Ian Colbert, Ihab Amer, Mehdi Saeedi, Gabor Sines, Nandita Vijaykumar
DPWatch: A Framework For Hardware-Based Differential Privacy Guarantees
IEEE Computer Architecture Letters (**IEEE CAL**), 2025.
20. Yintao He, Haiyu Mao, **Christina Giannoula**, Mohammad Sadrosadati, Juan Gomez Luna, Huawei Li, Xiaowei Li, Ying Wang, Onur Mutlu
PAPI: Exploiting Dynamic Parallelism in LLM Decoding with a PIM-enabled Computing System
Proceedings of the International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), 2025.
19. Sankeerth Durvasula, Adrian Zhao, Fan Chen, Ruofan Liang, Pawan Kumar Sanjaya, Yushi Guan, **Christina Giannoula**, Nandita Vijaykumar
ARC: Warp-level Adaptive Atomic Reduction in GPUs to Accelerate Differentiable Rendering

Proceedings of the International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), 2025.

Officially artifact evaluated as available and functional.

18. Zhanda, Zhu, **Christina Giannoula**, Muralidhar Andoorvedu, Qidong Su, Karttikeya Mangalam, Bojian Zheng, Gennady Pekhimenko
Mist: Efficient Distributed Training of Large Language Models via Memory-Parallelism Co-Optimization
Proceedings of the European Conference on Computer Systems (**EuroSys**), 2025.
Officially artifact evaluated as available, functional and reproducible.
17. Enrique Torres Sanchez, Milos Nikolic, Kareem Ibrahim, Alberto Delmas Lascorz, Ali Hadi Zadeh, Jiahui Wang, Ameer Abdelhadi, Mostafa Mahmoud, **Christina Giannoula**, Andreas Moshovos
Every Bit Matters: A Hardware/Software Approach for Enabling More Powerful Machine Learning Models
Proceedings of the Asilomar Conference on Signals, Systems, and Computers (**ACSSC**), 2024.
16. Cheng Chen, **Christina Giannoula**, Andreas Moshovos
Regular-Low-Bitwidth Floating Point Quantization for Efficient and High-Quality Diffusion Models
Proceedings of the IEEE International Symposium on Workload Characterization (**IISWC**), 2024.
15. Baorun Mu, **Christina Giannoula**, Shang Wang, Gennady Pekhimenko
Sylva: Sparse Embedded Adapters via Hierarchical Approximate Second-Order Information
Proceedings of the International Conference on Supercomputing (**ICS**), 2024.
14. Yubo Gao, Maryam Haghighifam, **Christina Giannoula**, Renbo Tu, Gennady Pekhimenko, Nandita Vijaykumar
Proteus: Preserving Model Confidentiality during Graph Optimizations
Proceedings of the 7th Annual Conference on Machine Learning and Systems (**MLSys**), 2024.
13. Alberto Delmas Lascorz, Mostafa Mahmoud, Ali Hadi Zadeh, Milos Nikolic, Kareem Ibrahim, **Christina Giannoula**, Ameer Abdelhadi, Andreas Moshovos
Atalanta: A Bit is Worth a "Thousand" Tensor Values
Proceedings of the International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), 2024.
12. Eugene Sha, Andy Liu, Kareem Ibrahim, Mostafa Mahmoud, **Christina Giannoula**, Ameer Abdelhadi, Andreas Moshovos
Marple: Scalable Spike Sorting for Untethered Brain-Machine Interfacing
Proceedings of the International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), 2024.
11. Ivan Fernandez, **Christina Giannoula**, Aditya Manglik, Ricardo Quislan, Nika Mansouri Ghiasi, Juan Gómez-Luna, Eladio Gutiérrez, Oscar Plata, Onur Mutlu
MATSA: An MRAM-based Energy-Efficient Accelerator for Time Series Analysis
IEEE Access, 2024.
10. Jiacheng Yang, **Christina Giannoula**, Jun Wu, Mostafa Elhoushi, James Gleeson, Gennady Pekhimenko
Minuet: Accelerating 3D Sparse Convolutions on GPUs
Proceedings of the European Conference on Computer Systems (**EuroSys**), 2024
Officially artifact evaluated as available, functional and reproducible.

9. **Christina Giannoula**, Kailong Huang, Jonathan Tang, Nectarios Koziris, Georgios Goumas, Zeshan Chishti, Nandita Vijaykumar
DaeMon: Architectural Support for Efficient Data Movement in Fully Disaggregated Systems
Proceedings of the ACM International Conference on Measurement and Modeling of Computer Systems (**SIGMETRICS & POMACS**), 2023.
8. **Christina Giannoula**, Athanasios Peppas, Georgios Goumas, Nectarios Koziris
High-Performance and Balanced Parallel Graph Coloring on Multicore Platforms
Journal of Supercomputing, 2022
7. Juan Gómez-Luna, Izzat El Hajj, Ivan Fernandez, **Christina Giannoula**, Geraldo F. Oliveira, Onur Mutlu
Benchmarking a New Paradigm: Experimental Analysis and Characterization of a Real Processing-in-Memory System
IEEE Access, 2022.
6. **Christina Giannoula**, Ivan Fernandez, Juan Gómez-Luna, Nectarios Koziris, Georgios Goumas, Onur Mutlu
SparseP: Towards Efficient Sparse Matrix Vector Multiplication on Real Processing-In-Memory Architectures
Proceedings of the ACM International Conference on Measurement and Modeling of Computer Systems (**SIGMETRICS & POMACS**), 2022.
5. **Christina Giannoula**, Nandita Vijaykumar, Nikela Papadopoulou, Vasileios Karakostas, Ivan Fernandez, Juan Gómez-Luna, Lois Orosa, Nectarios Koziris, Georgios Goumas, Onur Mutlu
SynCron: Efficient Synchronization Support for Near-Data-Processing Architectures
Proceedings of the International Symposium on High-Performance Computer Architecture (**HPCA**), 2021.
HiPEAC Awards 2021.
4. Ivan Fernandez, Richardo Quislan, **Christina Giannoula**, Mohammed Alser, Juan Gómez-Luna, Eladio Gutiérrez, Oscar Plata, Onur Mutlu
NATSA: A Near-Data Processing Accelerator for Time Series Analysis
Proceedings of the IEEE International Conference on Computer Design (**ICCD**), 2020.
3. Konstantinos Kanellopoulos, Nandita Vijaykumar, **Christina Giannoula**, Roknoddin Azizi, Skanda Koppula, Nika Mansouri Ghiasi, Taha Shahroodi, Juan Gómez-Luna, Onur Mutlu
SMASH: Co-designing Software Compression and Hardware-Accelerated Indexing for Efficient Sparse Matrix Operations
Proceedings of the 52nd International Symposium on Microarchitecture (**MICRO**), 2019.
HiPEAC Awards 2019.
2. **Christina Giannoula***, Foteini Strati*, Dimitrios Siakavaras, Georgios Goumas, Nectarios Koziris
* Joint first authors
An Adaptive Concurrent Priority Queue for NUMA Architectures
Proceedings of the 16th ACM International Conference on Computing Frontiers (**CF**), 2019
1. **Christina Giannoula**, Georgios Goumas, Nectarios Koziris
Combining Hardware Transactional Memory with Read Copy Update to Speed Up Graph Coloring on Multicore Platforms
Proceedings of the ISC High Performance (**ISC HPC**), 2018.
2nd Place Winner at PACT ACM Student Research Competition.

PREPRINT PUBLICATIONS

2. Qidong Su, **Christina Giannoula**, Gennady Pekhimenko
The Synergy of Speculative Decoding and Batching in Serving Large Language Models
[arXiv 2023](#)
1. Nika Mansouri Ghiasi, Mohammad Sadrosadati, Geraldo Oliveira, Konstantinos Kanellopoulos, Rachata Ausavarungnirun, Juan Gómez Luna, Aditya Manglik, João Ferreira, Jeremie Kim, **Christina Giannoula**, Nandita Vijaykumar, Jisung Park, Onur Mutlu
RevaMp3D: Architecting the Processor Core and Cache Hierarchy for Systems with Monolithically-Integrated Logic and Memory
[arXiv 2022](#)

RESEARCH PROPOSAL WRITING EXPERIENCE

- NSERC Discovery Research Proposal, \$265K fund for 5 years
Proposal: Computer Systems for Efficient and Scalable Machine Learning
PI: Prof. Gennady Pekhimenko, April 2024
- Mitacs Accelerate Research Proposal, \$500K fund for 3 years
Proposal: System-Level Optimizations for Emerging Deep Learning Models on Modern Computing Systems
PI: Prof. Gennady Pekhimenko, January 2025

TEACHING EXPERIENCE

Principal Instructor and Lecturer:

- ◇ Computer Organization (UofT, CSCB58, Undergraduate) Winter 2024
 - Co-teaching with Prof. Nandita Vijaykumar
 - 78 Undergraduate Students
 - 5 Teaching Assistants (TAs)
- ◇ Computer Organization (UofT, CSCB58, Undergraduate) Summer 2023
 - 49 Undergraduate Students
 - 2 Teaching Assistants (TAs)
- ◇ Advanced Computing Systems: Code Optimization Techniques For Multiprocessor Architectures (NTUA, ECE Graduate-632, PhD-level) Fall 2022
 - Co-teaching with Prof. Georgios Goumas
 - 6 PhD Students

Teaching Assistant (TA):

- ◇ Operating Systems (NTUA, ECE-3.4.3136.6) Spring 2022
- ◇ Parallel Processing Systems (NTUA, ECE-3.4.3257.9) Fall 2021
- ◇ Operating Systems (NTUA, ECE-3.4.3136.6) Spring 2021
- ◇ Parallel Processing Systems (NTUA, ECE-3.4.3257.9) Fall 2020
- ◇ Operating Systems (NTUA, ECE-3.4.3136.6) Spring 2020
- ◇ Advanced Topics in Computer Architecture (NTUA, ECE-3.4.3352.8) Spring 2020
- ◇ Parallel Processing Systems (NTUA, ECE-3.4.3257.9) Fall 2017
- ◇ Operating Systems (NTUA, ECE-3.4.3136.6) Fall 2017
- ◇ Computer Architecture (NTUA, ECE-3.4.3357.5) Fall 2017
- ◇ Operating Systems (NTUA, ECE-3.4.3136.6) Fall 2016
- ◇ Operating Systems (NTUA, ECE-3.4.3136.6) Fall 2015

SUPERVISED STUDENTS & MENTEES

Current Mentees:

4. **Dionysios Adamopoulos**, *Accelerating Point Cloud Networks on GPUs*, Diploma Thesis Research, ECE NTUA. 2025–Present
3. **Shixin Zhao**, *Accelerating Mixture of Experts Leveraging Processing-In-Memory*, PhD Research, Institute of Computing Technology, Chinese Academy of Sciences & ETH Zurich. 2025–Present
2. **Pawan Kumar Sanjaya**, *Guaranteeing Differentially Private Training at the Hardware Level*, PhD Research, University of Toronto. 2024–Present
1. **Peiming Yang**, *Accelerating Emerging Machine Learning Models on Real Processing-In-Memory Architectures*, PhD Research, University of Toronto. 2023–Present

Mentees Alumni:

17. **Yintao He**, *Exploiting Dynamic Parallelism in LLM Decoding with a PIM-enabled Computing System*, PhD Research, Institute of Computing Technology, Chinese Academy of Sciences & ETH Zürich. First position: Postdoctoral Researcher at Duke University. 2024–2025
16. **Pavel Golikov**, *Hardware Acceleration of Low-Bitwidth Quantized Large Language Models*, PhD Research, University of Toronto. 2023–2025
15. **Qidong Su**, *Seesaw: High-throughput LLM Inference via Model Re-Sharding, & The Synergy of Speculative Decoding and Batching in Serving Large Language Models*, PhD Research, University of Toronto. 2023–2025
14. **Zhanda Zhu**, *Mist: Efficient Distributed Training of Large Language Models via Memory-Parallelism Co-Optimization*, PhD Research, University of Toronto. 2023–2025
13. **Yubo Gao**, *Proteus: Preserving Model Confidentiality during Graph Optimizations*, PhD Research, University of Toronto. 2023–2025
12. **Jiacheng Yang**, *Minuet: Accelerating 3D Sparse Convolutions on Modern GPUs*, PhD Research, University of Toronto. 2023–2025
11. **Cheng Chen**, *Regular-Low-Bitwidth Floating Point Quantization for Efficient and High-Quality Diffusion Models*, MSc Research, University of Toronto. First position: Engineer at Amazon. 2024–2025
10. **Baorun Mu**, *Sylva: Sparse Embedded Adapters via Hierarchical Approximate Second-Order Information*, PhD Research, University of Toronto. 2023–2025
9. **Marzieh Barkhordar**, *Sparse Linear Algebra for Graph Processing on Processing-In-Memory Architectures*, PhD Research, Simon Fraser University. 2023–2025
8. **Lukas Zink**, *Towards Throughput-oriented Sparse Matrix Vector Multiplication on a Processing-in-Memory System*, MSc Research, ETH Zürich. 2023–2024
7. **Konstantinos Papadopoulos**, *Evaluating Microarchitectural Hardware Fault Detection for Application-Specific Requirements*, Diploma Thesis Research, ECE NTUA. First position: PhD at Northwestern University. 2023–2025
6. **Yu Xin Li**, *Quantization Methods for Graph Neural Networks*, Undergraduate Research, University of Toronto. 2023–2024
5. **Thrasyvoulos-Fivos Iliadis**, *SecureBoost: Accelerating the Secure Boot Process in Modern Microcontrollers*, Diploma Thesis Research, ECE NTUA. First position: PhD at NTUA. 2021–2023
4. **Jonathan Tang**, *DaeMon: Architectural Support for Efficient Data Movement in Disaggregated Systems*, Undergraduate Research, University of Toronto. 2020–2022
3. **Kailong Huang**, *DaeMon: Architectural Support for Efficient Data Movement in Disaggregated Systems*, Undergraduate Research, University of Toronto. First position: Engineer at AMD. 2020–2022
2. **Athanasios Peppas**, *High-Performance and Balanced Parallel Graph Coloring on Multicore Platforms*, Diploma Thesis Research, ECE NTUA. 2021–2022
1. **Foteini Strati**, *High-Performance Concurrent Priority Queues for NUMA Architectures*, Diploma Thesis Research, ECE NTUA. First position: PhD at ETH Zürich. 2018–2019

OPEN-SOURCE PROJECTS

- ◇ **Mist:** <https://github.com/CentML/Mist.git>
“Mist: Efficient Distributed Training of Large Language Models via Memory-Parallelism Co-Optimization”, EuroSys 2025
- ◇ **PyGim:** <https://github.com/CMU-SAFARI/PyGim.git>
“PyGim : An Efficient Graph Neural Network Library for Real Processing-In-Memory Architectures”, ACM Sigmetrics/POMACS 2025
- ◇ **ARC:** <https://github.com/Accelsnow/gaussian-splatting-distwar.git>
“ARC: Warp-level Adaptive Atomic Reduction in GPUs to Accelerate Differentiable Rendering”, ASPLOS 2025
- ◇ **Sylva:** <https://github.com/CentML/Sylva.git>
“Sylva: Sparse Embedded Adapters via Hierarchical Approximate Second-Order Information”, ICS 2024
- ◇ **Proteus:** <https://github.com/proteus-mlsys24/mlsys24-artifact.git>
“Proteus: Preserving Model Confidentiality during Graph Optimizations”, MLSys 2024
- ◇ **Minuet:** <https://github.com/UofT-EcoSystem/Minuet.git>
“Minuet: Accelerating 3D Sparse Convolutions on GPUs”, EuroSys 2024
- ◇ **SparseP:** <https://github.com/CMU-SAFARI/SparseP.git>
“SparseP: Towards Efficient Sparse Matrix Vector Multiplication on Real Processing-In-Memory Architectures ”, ACM Sigmetrics/POMACS 2022
- ◇ **Prim:** <https://github.com/CMU-SAFARI/prim-benchmarks.git>
“Benchmarking a New Paradigm: Experimental Analysis and Characterization of a Real Processing-in-Memory System ”, IEEE Access 2022
- ◇ **NATSA:** <https://github.com/CMU-SAFARI/NATSA.git>
“NATSA: A Near-Data Processing Accelerator for Time Series Analysis”, ICCD 2020
- ◇ **SMASH:** <https://github.com/CMU-SAFARI/SMASH.git>
“SMASH: Co-designing Software Compression and Hardware-Accelerated Indexing for Efficient Sparse Matrix Operations”, MICRO 2019
- ◇ **ColorTM:** <https://github.com/cgiannoula/ColorTM.git>
“Combining HTM with RCU to Speed up Graph Coloring on Multicore Platforms”, ISC HPC 2018
“High-Performance and Balanced Parallel Graph Coloring on Multicore Platforms”, Journal of Supercomputing 2022

PATENTS

- ◇ **System and Method for Performing High-Throughput Inference Via Model Resharding**, US Patent Application Filed January 2025
- ◇ **System and Method for Fine Tuning Large Language Models (LLMs) Using Sparse Embedded Adapters Via Hierarchical Approximate Second-Order Information**, US Patent Application Filed June 2024
- ◇ **System and Method for Efficient Large Scale Distributed Training**, US Patent Application Filed August 2024
- ◇ **System and Computer-Implemented Method for Preserving Model Confidentiality During Graph Optimizations**, US Patent Application Filed October 2024

INVITED TALKS

Hardware-Efficient Computational Kernels

- ◇ The Cornell, Maryland, Max Planck Pre-doctoral Research School in Computer Science (CMMRS 2025),
Invited Talk July 2025

<i>PyGim: An Efficient Graph Neural Network Library for Real Processing-In-Memory Architectures</i>	
◇ Memory-Centric Computing Systems Workshop (with ASPLOS 2025), Invited Talk	April 2025
<i>System Software and Libraries for Sparse Computational Kernels in PIM Architectures</i>	
◇ Memory-Centric Computing Systems Tutorial (with MICRO 2024), Invited Talk	November 2024
<i>Let's Go Sparse: Towards Sparse-Aware Computing System Design</i>	
◇ Meta, Sunnyvale, Virtually, Invited Talk	August 2024
◇ University of Illinois Urbana-Champaign, Invited Talk	June 2024
◇ Northwestern University, Invited Talk	June 2024
◇ University of Michigan, Invited Talk	June 2024
◇ University of Edinburgh, Virtually, Invited Talk	May 2024
◇ University of California San Diego, Invited Talk	May 2024
◇ Qualcomm Research Center, San Diego	May 2024
<i>Accelerating 3D Sparse Convolutions on GPUs</i>	
◇ Discovery AI Forum organized by LG Electronics, Virtually, Invited Talk	December 2023
<i>Accelerating Deep Learning Training on Edge Devices</i>	
◇ Discovery AI Forum organized by LG Electronics, Virtually, Invited Talk	December 2023
<i>DaeMon: Architectural Support for Efficient Data Movement in Fully Disaggregated Systems</i>	
◇ AMD California, Virtually, Invited Talk	July 2023
◇ ETH Zürich, Virtually, Seminar Talk	July 2023
◇ SIGMETRICS 2023, Orlando, USA, Conference Talk	June 2023
◇ HCM Workshop (in conjunction with HPCA 2023), Montreal, Canada, Invited Talk	February 2023
<i>Accelerating Irregular Applications via Efficient Synchronization and Data Access Techniques</i>	
◇ ETH Zürich, Virtually, Seminar Talk	November 2022
◇ ECE NTUA, Athens, Greece, PhD Defense Talk	September 2022
<i>SparseP: Towards Efficient Sparse Matrix Vector Multiplication on Real Processing-In-Memory Architectures</i>	
◇ Real PIM Tutorial (in conjunction with ISCA 2023) Orlando, USA, Invited Talk	June 2023
◇ Real PIM Tutorial (in conjunction with HPCA 2023) Montreal, Canada, Invited Talk	February 2023
◇ ISVLSI 2022, Virtually, Conference Talk	July 2022
◇ SIGMETRICS 2022, Mumbai, India, Conference Talk	June 2022
◇ ETH Zürich, Virtually, Processing-In-Memory Course Lecture	May 2022
◇ Huawei Research & University of Edinburgh, Virtually	May 2022
<i>SynCron: Efficient Synchronization Support for Near-Data Processing Architectures</i>	
◇ ETH Zürich, Virtually, Ramulator Course Lecture	May 2022
◇ AMD Research, Virtually	February 2022
◇ ETH Zürich, Virtually, Cutting-Edge Research Lecture	December 2021
◇ ETH Zürich, Virtually, Processing-In-Memory Course Lecture	December 2021
◇ ETH Zürich, Virtually, Seminar Talk	September 2021
◇ HPCA 2021, Virtually, Conference Talk	March 2021
◇ Intel Labs, Virtually	April 2020
<i>An Adaptive Concurrent Priority Queue for NUMA Architectures</i>	
◇ CF 2019, Sardinia, Italy	April 2019
<i>Employing Hardware Transactional Memory to Design a Balanced Graph Coloring Algorithm</i>	
◇ PACT 2018, Limassol, Cyprus	November 2018
<i>Combining HTM with RCU to Speed Up Graph Coloring on Multicore Platforms</i>	

- ◇ ETH Zürich, Zürich, Switzerland
- ◇ ISC High Performance 2018, Frankfurt, Germany

January 2019
June 2018

SERVICE

Organization and Service Committees:

- ◇ Social Media Editor, ACM SIGMICRO September 2023 - Present
- ◇ Publicity Co-Chair, EuroSys 2026
- ◇ Program Chair & Organizing Committee, 4th Workshop on Heterogeneous Composable and Disaggregated Systems (HCDS) (in conjunction with ASPLOS) 2024, 2025
- ◇ Organizing Committee, SPICE: Sparsity, the Key Ingredient from HPC to Efficient LLMs (in conjunction with MICRO 2025)
- ◇ Organizing Committee, Tiny Titans: The next wave of On-Device Learning for Foundational Models (TTODLer-FM) (in conjunction with ICML 2025)
- ◇ Publicity Chair, ICCD 2024
- ◇ Research Adjudication Committee, Vector Institute AI Scholarship 2024
- ◇ Publicity Chair, HPCA 2024
- ◇ Publicity & Social Media Chair, MICRO 2023
- ◇ Publicity Chair, CompSys 2023 (in conjunction with IPDPS 2023)

Program Committee (PC) Member:

- ◇ SIGMETRICS 2026
- ◇ HPCA 2024, 2026
- ◇ PACT 2024
- ◇ MLSys 2024
- ◇ ASPLOS 2024
- ◇ Memory-Centric Computing Workshop Workshop 2025 (in conjunction with ASPLOS 2025)
- ◇ EuroMLSys 2024 (in conjunction with EuroSys 2024)
- ◇ Young Architect Workshop (YArch) 2024 (in conjunction with ASPLOS 2024)
- ◇ AsHES (in conjunction with IPDPS 2024)
- ◇ DATE 2023
- ◇ FastPath 2022 (in conjunction with MICRO 2022)
- ◇ Shadow Program Committee Member, EuroSys 2021

External Reviewer Committee (ERC) Member:

- ◇ ASPLOS 2025, 2026
- ◇ ISCA 2024

Journal Referee:

- ◇ ACM TACO 2023
- ◇ Special Issue on Near-Memory and In-Memory, IEEE Transactions on Computers (TC) 2023

Artifact Evaluation Committee Member:

- ◇ MICRO 2022